

Coulomb-Barrier-Free Neutron Production via Proton–Electron Capture in Dense Hydrogen Gas

with Lithium-6 Energy Extraction

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Abstract

We propose a theoretical mechanism for neutron production and energy extraction that entirely avoids the Coulomb barrier. A free proton injected into dense hydrogen gas interacts with a bound electron via electrostatic attraction, producing a neutron through inverse beta decay ($p + e^- \rightarrow n + \nu_e$). The liberated hydrogen nucleus then becomes a free proton, capable of repeating the reaction on a neighboring hydrogen atom, forming a self-propagating chain. Produced neutrons are captured by Lithium-6, releasing 4.78 MeV per event via $n + {}^6\text{Li} \rightarrow {}^4\text{He} + {}^3\text{H} + 4.78\text{ MeV}$. The net energy output per cycle is 3.487 MeV. The mechanism requires no Coulomb barrier crossing and represents a novel conceptual pathway for low-energy neutron-driven nuclear energy.

Keywords: inverse beta decay, Coulomb barrier, chain reaction, neutron capture, lithium-6, LENR, hydrogen gas

1 Introduction

The Coulomb barrier is the primary obstacle in achieving controlled nuclear reactions at low energies. In conventional fusion, two positively charged nuclei must overcome a repulsive potential on the order of 1 MeV at nuclear distances, requiring extreme temperatures or pressures to proceed at useful rates.

We propose a fundamentally different approach: instead of fusing two positive nuclei, we exploit the *attractive* interaction between a free proton and a bound electron in a hydrogen atom. Since the proton and electron carry opposite charges, no repulsive Coulomb barrier exists. The resulting neutrons, being electrically neutral, can penetrate target nuclei without any Coulomb barrier, making Lithium-6 an ideal energy extraction target.

2 Reaction Mechanism

2.1 Step 1: Neutron Production

A free proton with kinetic energy $E_k \geq 1.293\text{ MeV}$ moves through dense hydrogen gas. Upon approaching a hydrogen atom, the proton is attracted to the bound electron:

$$F = -\frac{e^2}{4\pi\epsilon_0 r^2} \quad (\text{attractive, no Coulomb barrier}) \quad (1)$$

When the proton and electron interact at sufficiently close range, inverse beta decay proceeds:

$$p + e^- \rightarrow n + \nu_e \quad \Delta mc^2 = -1.293 \text{ MeV} \quad (2)$$

2.2 Step 2: Chain Reaction

After Eq. (1), the original hydrogen nucleus is released as a new free proton. This proton repeats the process on the next hydrogen atom, forming a chain:

$$p^{(1)} + \text{H}_1 \rightarrow n_1 + \nu_e + p^{(2)} \quad (\text{C1})$$

$$p^{(2)} + \text{H}_2 \rightarrow n_2 + \nu_e + p^{(3)} \quad (\text{C2})$$

$\vdots$

$$p^{(i)} + \text{H}_i \rightarrow n_i + \nu_e + p^{(i+1)} \quad (\text{Ci})$$

Each step consumes one hydrogen atom and produces one neutron and one free proton that sustains the chain.

2.3 Step 3: Energy Extraction via Lithium-6

Produced neutrons are directed into a Lithium-6 blanket:

$$n + {}^6\text{Li} \rightarrow {}^4\text{He} + {}^3\text{H} + 4.78 \text{ MeV} \quad (3)$$

This reaction is exothermic, emits no gamma radiation, and the kinetic energy of products thermalizes into usable heat.

3 Energy Balance

Table 1: Energy budget per complete reaction cycle

Process	Energy	Direction
$p + e^- \rightarrow n + \nu_e$	1.293 MeV	Input
$n + {}^6\text{Li} \rightarrow {}^4\text{He} + {}^3\text{H}$	4.780 MeV	Output
Net per cycle	3.487 MeV	Output

$$\boxed{E_{\text{net}} = 4.780 - 1.293 = 3.487 \text{ MeV per neutron}} \quad (4)$$

3.1 Chain Sustainability

The chain sustains itself when each liberated proton retains sufficient energy to trigger the next capture. Ionization losses over one mean free path λ must not reduce E_k below 1.293 MeV:

$$E_k^{(i+1)} = E_k^{(i)} - 1.293 \text{ MeV} - E_{\text{loss}}(\lambda) \geq 1.293 \text{ MeV} \quad (5)$$

The mean free path at pressure P and temperature T is:

$$\lambda = \frac{k_B T}{\sqrt{2} \pi d_H^2 P} \quad (6)$$

where $d_H \approx 120$ pm. At $P = 100$ atm, $\lambda \approx 2$ nm, minimizing ionization losses and favoring chain propagation.

4 Comparison with Conventional Approaches

Table 2: Comparison with related nuclear energy mechanisms

Feature	D–T Fusion	Fission	This Work
Coulomb barrier	High	None (n)	None
Fuel	D, T	U-235	H ₂ , Li-6
Chain mechanism	No	Yes	Yes
Radioactive waste	Low	High	Low
Energy per event	17.6 MeV	200 MeV	3.487 MeV

5 Open Challenges

1. **Energy threshold.** The initiating proton must carry $E_k \geq 1.293$ MeV. A particle accelerator or high-energy ion source is required to start the chain.
2. **Ionization losses.** The free proton loses energy traversing hydrogen gas. High pressure operation ($P > 50$ atm) is necessary to keep λ short enough to sustain the chain.
3. **Neutrino energy loss.** Each event emits a neutrino carrying ~ 0.511 MeV that escapes unrecovered. This is included in the energy balance.
4. **Neutron capture efficiency.** Neutrons must reach the Li-6 blanket before escaping or thermalizing elsewhere. Neutron reflectors and optimized geometry are required.

6 Proposed Experimental Test

1. Compress hydrogen gas to $P > 50$ atm in a chamber surrounded by a Li-6 neutron capture blanket.

2. Inject a proton beam at $E_k = 5\text{--}20$ MeV into the gas.
3. Measure neutron flux at the Li-6 blanket versus hydrogen pressure. A superlinear dependence on pressure would indicate chain propagation.
4. Measure heat output via calorimetry. Net output exceeding beam energy input would confirm the energy-positive cycle.

7 Conclusion

We have proposed a Coulomb-barrier-free neutron production mechanism based on proton–electron capture in dense hydrogen gas, with energy extraction via Lithium-6 neutron capture. The key features are:

- No Coulomb barrier (proton–electron attraction).
- Self-propagating chain reaction via liberated protons.
- Net energy output: 3.487 MeV per reaction cycle.
- Abundant, low-cost fuel: hydrogen and Lithium-6.

Whether the chain can sustain itself against ionization losses is the central open question, directly testable with existing accelerator technology.

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